B Review of linear algebraic systems, eigenvalues and eigenvectors[26]

[26] See the lecture notes from MTH1002, part 2

B.1 Linear algebraic systems

Suppose we want to solve a set of n linear algebraic equations in n variables

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n\,. \end{aligned} \tag{B.1}$$

We can write this as the matrix problem

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}\,, \tag{B.2}$$

where $\boldsymbol{A}$ is an $n \times n$ matrix and $\boldsymbol{b}$ is a column vector, both of which are known. We seek the solutions $\boldsymbol{x}$.

Definition B.1 If $\boldsymbol{b} = \boldsymbol{0}$ the linear system (B.2) is said to be **homogeneous**. If $\boldsymbol{b} \neq \boldsymbol{0}$ the system is **nonhomogeneous**.

If $\boldsymbol{A}$ is non-singular (i.e. the determinant of $\boldsymbol{A}$ is non-zero) then the inverse $\boldsymbol{A}^{-1}$ exists and the solution of (B.2) is found by multiplying each side by $\boldsymbol{A}^{-1}$ to obtain

$$\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}\,.$$

It is clear that the homogeneous system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ then only has the **trivial solution** $\boldsymbol{x} = \boldsymbol{0}$.

If $\boldsymbol{A}$ is singular, $\boldsymbol{A}^{-1}$ does not exist. In this case, it can be shown that the homogeneous system

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0} \tag{B.3}$$

has infinitely many non-zero solutions, in addition to the trivial solution. The nonhomogeneous system has no solutions unless the inner product

$$(\boldsymbol{b}, \boldsymbol{y}) = 0 \tag{B.4}$$

for all $\boldsymbol{y}$ satisfying

$$\bar{\boldsymbol{A}}^T\boldsymbol{y} = \boldsymbol{0}$$

(recall that $\bar{\boldsymbol{A}}^T$ is the adjoint of $\boldsymbol{A}$; see Appendix A). If condition (B.4) is satisfied then the nonhomogeneous problem has infinitely many solutions. Each of these solutions has the form

$$\boldsymbol{x} = \boldsymbol{x}^{(0)} + \boldsymbol{\xi}\,,$$

where $\boldsymbol{x}^{(0)}$ is a particular solution of (B.2) and $\boldsymbol{\xi}$ is any solution of the homogeneous system (B.3).

B.2 Eigenvalues and eigenvectors

We seek non-zero solutions $\boldsymbol{x}$ of the matrix problem

$$\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{x}\,, \tag{B.5}$$

where the $n \times n$ matrix $\boldsymbol{A}$ is given. Equivalently, this system can be written

$$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{x} = \boldsymbol{0}\,, \tag{B.6}$$

and this has non-zero solutions if and only if λ is chosen so that

$$|\boldsymbol{A} - \lambda \boldsymbol{I}| = 0\,. \tag{B.7}$$

Equation (B.7) is called the **characteristic equation** of the matrix $\boldsymbol{A}$.

Definition B.2 The roots of the characteristic equation (B.7) are called the **eigenvalues** of the matrix $\boldsymbol{A}$. The non-zero solutions $\boldsymbol{x}$ that are obtained by using each eigenvalue λ and solving (B.6) (or (B.5)) are called the **eigenvectors** corresponding to that eigenvalue.

Activity B.3 Show that if $\boldsymbol{x}$ is an eigenvector of $\boldsymbol{A}$ corresponding to eigenvalue λ, then for any non-zero scalar c, $c\boldsymbol{x}$ is also an eigenvector of $\boldsymbol{A}$ with the same eigenvalue.

Activity B.4 Find the eigenvalues λ and eigenvectors $\boldsymbol{x}$ of the matrix

$$\boldsymbol{A} = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix}. \tag{B.8}$$

Remember to check your answer by comparing $\boldsymbol{Ax}$ and $\lambda\boldsymbol{x}$.

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B.3 Linear dependence and independence of vectors

Definition B.5 A set of k vectors $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(k)}$ is said to be **linearly dependent** if there is a linear relationship among them, i.e. if there exists a set of (possibly complex) numbers $c_1, \ldots c_k$, not all zero, such that

$$c_1\boldsymbol{x}^{(1)} + \cdots + c_k\boldsymbol{x}^{(k)} = \boldsymbol{0}\,. \tag{B.9}$$

If the only set $c_1, \ldots c_k$ for which (B.9) is satisfied is $c_1 = c_2 = \cdots = c_k = 0$ then the vectors $\boldsymbol{x}^{(1)}, \ldots \boldsymbol{x}^{(k)}$ are said to be **linearly independent**.

Now consider a set of n vectors, $\boldsymbol{x}^{(j)}$ for $j = 1, 2, \ldots, n$, each of which has n components. Let the ith component of the vector $\boldsymbol{x}^{(j)}$ be denoted by $x_{ij} = x_i^{(j)}$. Define the matrix $\boldsymbol{X}$ to be the $n \times n$ matrix with entries x_{ij}, then equation (B.9) can be written as

$$\begin{pmatrix} x_1^{(1)}c_1 + \cdots + x_1^{(n)}c_n \\ \vdots \\ x_n^{(1)}c_1 + \cdots + x_n^{(n)}c_n \end{pmatrix} = \begin{pmatrix} x_{11}c_1 + \cdots + x_{1n}c_n \\ \vdots \\ x_{n1}c_1 + \cdots + x_{nn}c_n \end{pmatrix} = \boldsymbol{X}\boldsymbol{c} = \boldsymbol{0}\,. \tag{B.10}$$

If $|\boldsymbol{X}| \neq 0$ then we have said in §B.1 that the only solution of (B.10) is $\boldsymbol{c} = \boldsymbol{0}$. However, if $|\boldsymbol{X}| = 0$ then there are non-zero solutions. Thus, the set of vectors $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ is linearly independent if and only if $|\boldsymbol{X}| \neq 0$.

Activity B.6 Determine whether the vectors

$$\boldsymbol{x}^{(1)} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \boldsymbol{x}^{(2)} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \quad \boldsymbol{x}^{(3)} = \begin{pmatrix} -4 \\ 1 \\ -11 \end{pmatrix}$$

are linearly independent or linearly dependent. If they are linearly dependent, find a linear relationship between them.

Sometimes it is useful to think of the columns of a matrix $\boldsymbol{A}$ as vectors. These column vectors are linearly independent if and only if $|\boldsymbol{A}| \neq 0$. It can also be shown that if $\boldsymbol{C} = \boldsymbol{AB}$ then $|\boldsymbol{C}| = |\boldsymbol{A}||\boldsymbol{B}|$, and so if the columns of both $\boldsymbol{A}$ and $\boldsymbol{B}$ are linearly independent then the columns of $\boldsymbol{C}$ are also linearly independent.

We can extend the above ideas of linear independence to a set of vector functions $\boldsymbol{x}^{(1)}(t), \ldots, \boldsymbol{x}^{(k)}(t)$ defined on an interval $I : \alpha < t < \beta$. The vectors are linearly dependent on I if there exists a set of constants $c_1, \ldots, c_k$, not all zero, such that $c_1\boldsymbol{x}^{(1)}(t) + \cdots + c_k\boldsymbol{x}^{(k)}(t) = \boldsymbol{0}$ for all t in the interval I. Otherwise, the vectors are linearly independent.

Since the characteristic equation (B.7) is a polynomial of degree n in λ, there are n eigenvalues $\lambda_1, \ldots, \lambda_n$, some of which may be repeated. If a given eigenvalue appears m times it is said to have **multiplicity** m. Each eigenvalue has at least one associated eigenvector. An eigenvalue of multiplicity m can have q linearly independent eigenvectors, where $1 \leq q \leq m$.

Theorem B.7 Let the $n \times n$ matrix $\boldsymbol{A}$ have r different eigenvalues λ_j, for $j = 1, \ldots, r$, with $1 < r \leq n$, and let $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(r)}$ be the corresponding eigenvectors. Then then eigenvectors $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(r)}$ are linearly independent.

If $\boldsymbol{A}$ has a repeated eigenvalue then there may be fewer than n linearly independent eigenvectors. This leads to complications later on when we solve systems of differential equations (see §4.4.3).

B.4 Diagonalisation of a square matrix[27]

[27] This is new material, not contained in MTH1002

If the $n \times n$ matrix A has n (the maximal number) of linearly independent eigenvectors then a very useful way of solving a linear system is to transform the matrix into a diagonal matrix. The eigenvectors help us accomplish this.

Let $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ be the n linearly independent eigenvectors corresponding to the eigenvalues $\lambda_1, \ldots, \lambda_n$. Denote the components of each eigenvector with a subscript, for example

$$\boldsymbol{x}^{(1)} = \begin{pmatrix} x_1^{(1)} \\ x_2^{(1)} \\ \vdots \\ x_n^{(1)} \end{pmatrix} \tag{B.11}$$

and form the $n \times n$ matrix $\boldsymbol{P}$ whose columns are the n eigenvectors, i.e.

$$\boldsymbol{P} = \begin{pmatrix} x_1^{(1)} & x_1^{(2)} & \ldots & x_1^{(n)} \\ x_2^{(1)} & x_2^{(2)} & \ldots & x_2^{(n)} \\ \vdots & \vdots & & \vdots \\ x_n^{(1)} & x_n^{(2)} & \ldots & x_n^{(n)} \end{pmatrix}. \tag{B.12}$$

Since the columns of $\boldsymbol{P}$ are linearly independent vectors, we know that $|\boldsymbol{P}| \neq 0$ and hence $\boldsymbol{P}^{-1}$ exists.

Now, the columns of the matrix $\boldsymbol{AP}$ are the vectors $\boldsymbol{Ax}^{(1)}, \ldots, \boldsymbol{Ax}^{(n)}$. By the definition of an eigenvector and eigenvalue, $\boldsymbol{Ax}^{(k)} = \lambda_k \boldsymbol{x}^{(k)}$, and so we have

$$\boldsymbol{AP} = \begin{pmatrix} \lambda_1 x_1^{(1)} & \ldots & \lambda_n x_1^{(n)} \\ \vdots & & \vdots \\ \lambda_1 x_n^{(1)} & \ldots & \lambda_n x_n^{(n)} \end{pmatrix}. \tag{B.13}$$

However, if we form the $n \times n$ diagonal matrix $\boldsymbol{D}$ whose diagonal elements are the eigenvalues of $\boldsymbol{A}$, namely

$$\boldsymbol{D} = \begin{pmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{pmatrix}, \tag{B.14}$$

then

$$\begin{aligned} \boldsymbol{PD} &= \begin{pmatrix} x_1^{(1)} & x_1^{(2)} & \ldots & x_1^{(n)} \\ x_2^{(1)} & x_2^{(2)} & \ldots & x_2^{(n)} \\ \vdots & \vdots & & \vdots \\ x_n^{(1)} & x_n^{(2)} & \ldots & x_n^{(n)} \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{pmatrix} \\ &= \begin{pmatrix} \lambda_1 x_1^{(1)} & \ldots & \lambda_n x_1^{(n)} \\ \vdots & & \vdots \\ \lambda_1 x_n^{(1)} & \ldots & \lambda_n x_n^{(n)} \end{pmatrix}. \end{aligned} \tag{B.15}$$

Comparing (B.13) and (B.15) we see that

$$\boldsymbol{AP} = \boldsymbol{PD} \tag{B.16}$$

and we can pre-multiply by $\boldsymbol{P}^{-1}$ to obtain

$$\boldsymbol{P}^{-1}\boldsymbol{AP} = \boldsymbol{P}^{-1}\boldsymbol{PD} = \boldsymbol{D} \tag{B.17}$$

and equivalently (by post-multiplying (B.16) by $\boldsymbol{P}^{-1}$)

$$\boldsymbol{A} = \boldsymbol{PDP}^{-1}\,.$$

Thus, if the eigenvalues and eigenvectors of the matrix $\boldsymbol{A}$ are known, then we can transform $\boldsymbol{A}$ into the diagonal matrix $\boldsymbol{D}$. The transformation (B.17) is called the **diagonalisation** of $\boldsymbol{A}$.

It is noted that if the $n \times n$ matrix $\boldsymbol{A}$ has fewer than n linearly independent eigenvectors then there is no matrix $\boldsymbol{P}$ such that $\boldsymbol{P}^{-1}\boldsymbol{AP} = \boldsymbol{D}$. In this case we say that $\boldsymbol{A}$ is not diagonalisable. We do not consider this case further in this module.

Activity B.8 For the matrix $\boldsymbol{A}$ in Activity B.4, find $\boldsymbol{P}$ such that $\boldsymbol{P}^{-1}\boldsymbol{AP} = \boldsymbol{D}$, where $\boldsymbol{D}$ is a diagonal matrix. Confirm your choice by explicitly calculating $\boldsymbol{P}^{-1}\boldsymbol{AP}$.

B.4.1 Cramer's rule

A straightforward way of solving the linear system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 &= b_1, \\ a_{21}x_1 + a_{22}x_2 &= b_2, \end{aligned} \tag{B.18}$$

is to multiply the first equation by a_{22} and the second by a_{12} and then subtract the results to obtain

$$x_1 = \frac{b_1 a_{22} - b_2 a_{12}}{a_{11}a_{22} - a_{12}a_{21}}\,.$$

A similar procedure yields x_2. The formula can be written succinctly using determinants:

$$x_1 = \frac{\Delta_1}{\Delta}, \quad x_2 = \frac{\Delta_2}{\Delta}\,, \tag{B.19}$$

where $\Delta = |\boldsymbol{A}|$ when (B.18) is written as the system $\boldsymbol{Ax} = \boldsymbol{b}$, and Δ_1 and Δ_2 are obtained from Δ by replacing its first or second column with the vector $\boldsymbol{b}$, namely

$$\Delta = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \quad \Delta_1 = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}, \quad \Delta_2 = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}. \tag{B.20}$$

The rule is not limited to 2×2 systems. It can be extended to higher-order systems in the expected way.